LAB 2 FINAL REPORT

Agile Backlog Creation & Sprint Simulation in Jira

Project	Municipal Infrastructure Damage Reporting (MIDR)
Jira Board	MIDR board
Lab	Lab 2
Student Name	Hemanth v
Date	28 August 2026

1. Objective

The objective of this laboratory is to use Jira to convert functional requirements into Agile backlog items, simulate sprints, and analyse sprint progress.

2. Project Overview

The project used for this laboratory is the Municipal Infrastructure Damage Reporting (MIDR) system. The Jira backlog was organized using Epics and User Stories, with priorities and story points assigned before sprint execution.

3. Agile Backlog: Epics and User Stories

The Jira project contains the following functional themes and user stories visible in the completed sprint reports:

Epic / Functional Area	User Story	Priority	Story Points
Damage Report Submission	MIDR-4 — Submit Infrastructure Damage Report	High	5
Damage Report Submission	MIDR-5 — Upload Damage Photo	High	3
Location & Report Dispatch	MIDR-6 — Extract GPS Location from Photo	High	3
Location & Report Dispatch	MIDR-7 — Automatically Dispatch Damage Report	High	5
Engineer Report Management	MIDR-8 — Review and Update Damage Report Status	Medium	5
Engineer Report Management	MIDR-9 — Engineer Reviews Damage Report	High	3
Engineer Report Management	MIDR-10 — Update Damage Report Status	Medium	5

4. Story Point Assignments and Prioritization

Story points were assigned using the Fibonacci-style estimation approach described in the lab manual. The completed Sprint 1 report records a total of 21 story points, while Sprint 2 records 8 story points. The priorities visible in Jira were predominantly High, with Medium priority used for the status-management stories.

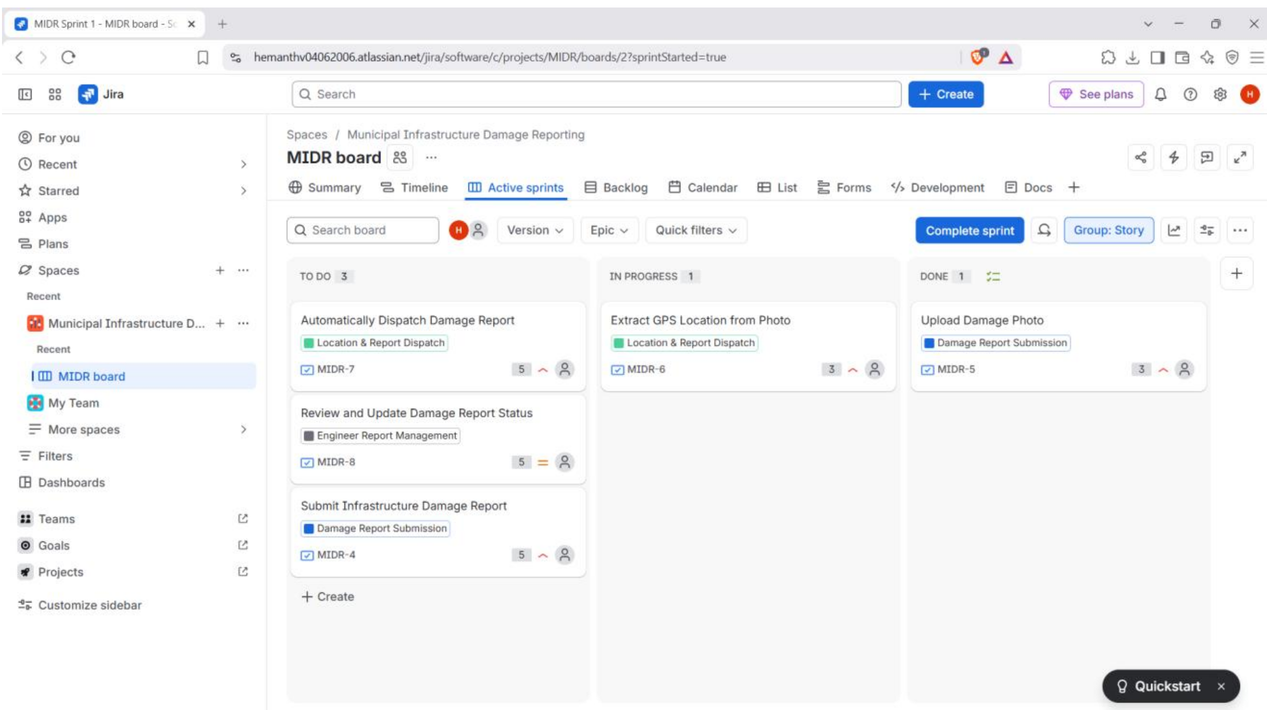
The lab manual explains that story points represent relative effort considering complexity, amount of work, risk, and uncertainty. Planning Poker is suggested as an estimation technique for team discussion.

Evidence placement: Use the Sprint Report screenshots below as evidence of the final story-point assignments.

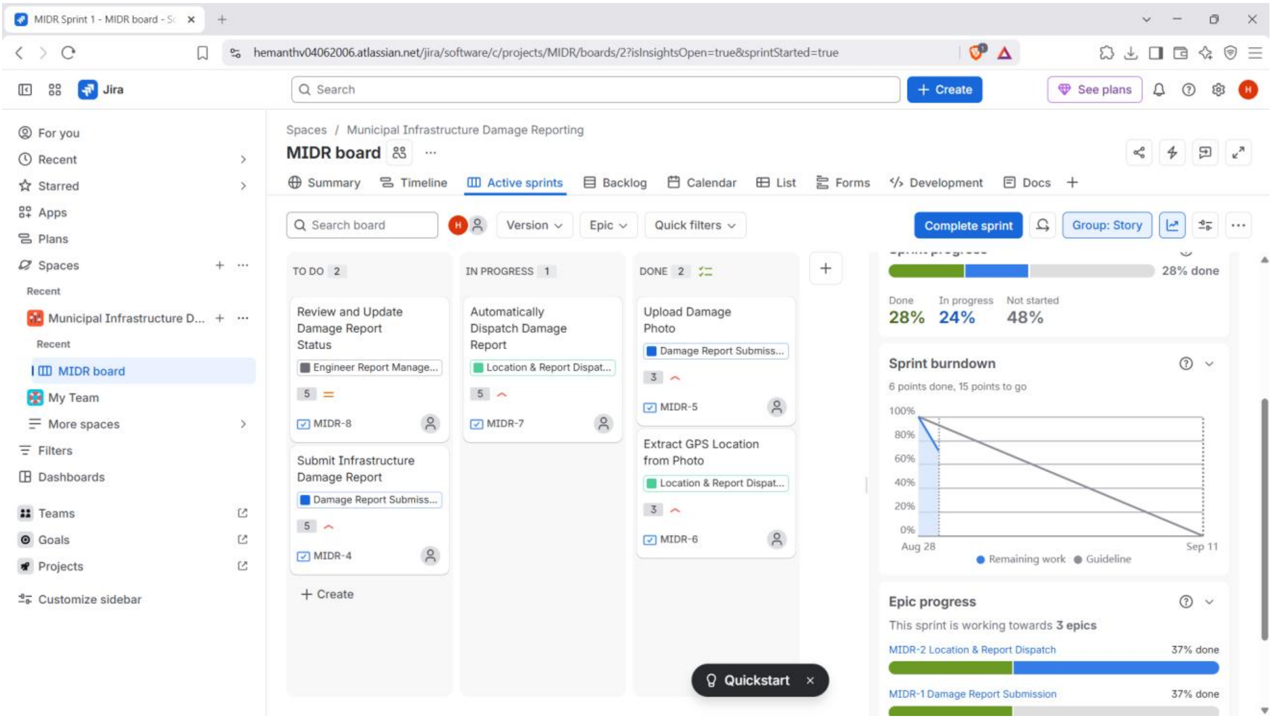
5. Sprint 1 Execution

Sprint 1 was executed on the MIDR Scrum board. The stories were progressed through the Jira workflow from To Do to In Progress and Done. The screenshots show the changing sprint state during the simulation.

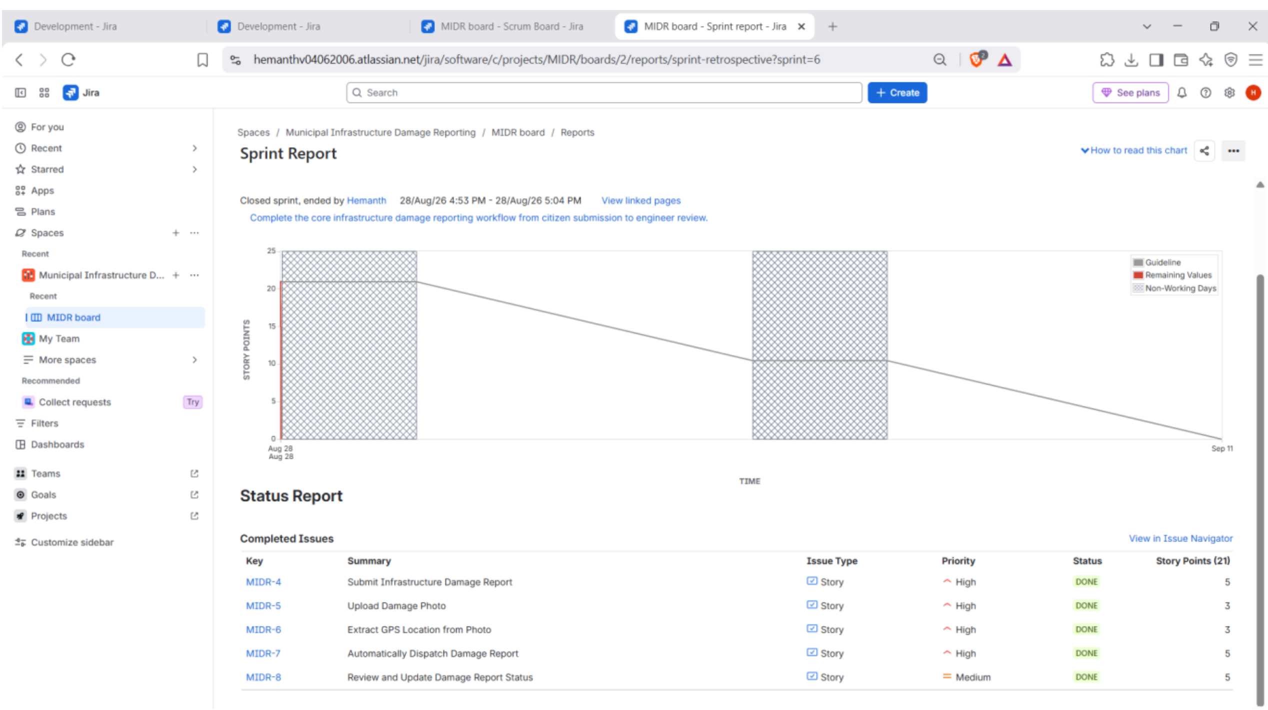
Screenshot 1 — Sprint 1 – Active Sprint board (initial progress)



Screenshot 2 — Sprint 1 – Active Sprint board (later progress)



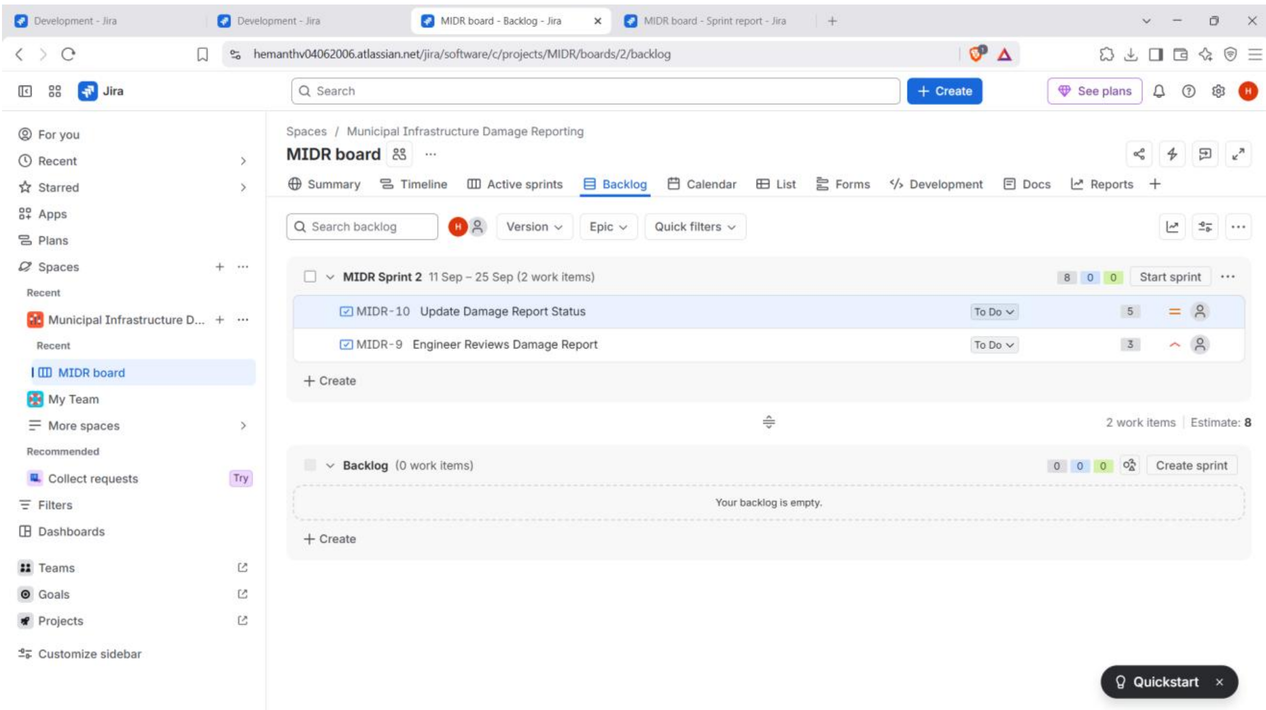
Screenshot 3 — Sprint 1 – Sprint Report



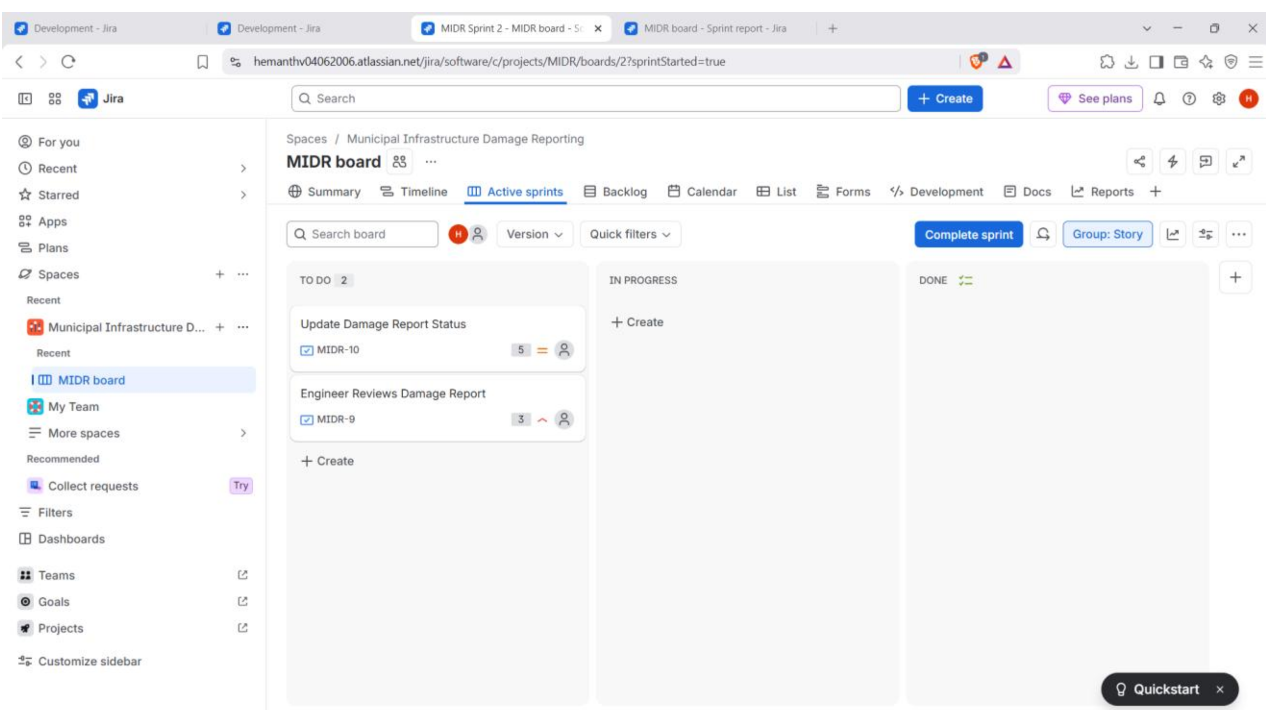
6. Sprint 2 Planning and Execution

After completing Sprint 1, the Jira backlog was used to create Sprint 2. Sprint 2 contained two work items: MIDR-9 Engineer Reviews Damage Report and MIDR-10 Update Damage Report Status, with an estimated total of 8 story points.

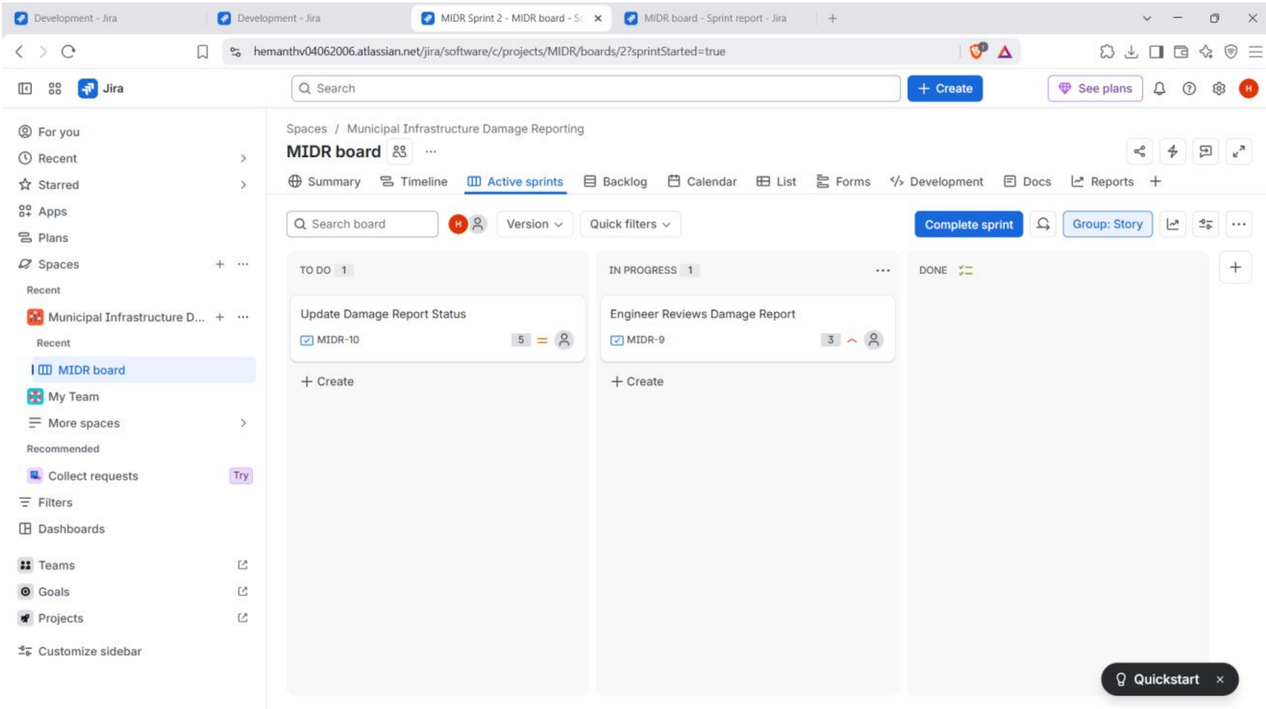
Screenshot 4 — Sprint 2 – Backlog before/at sprint setup



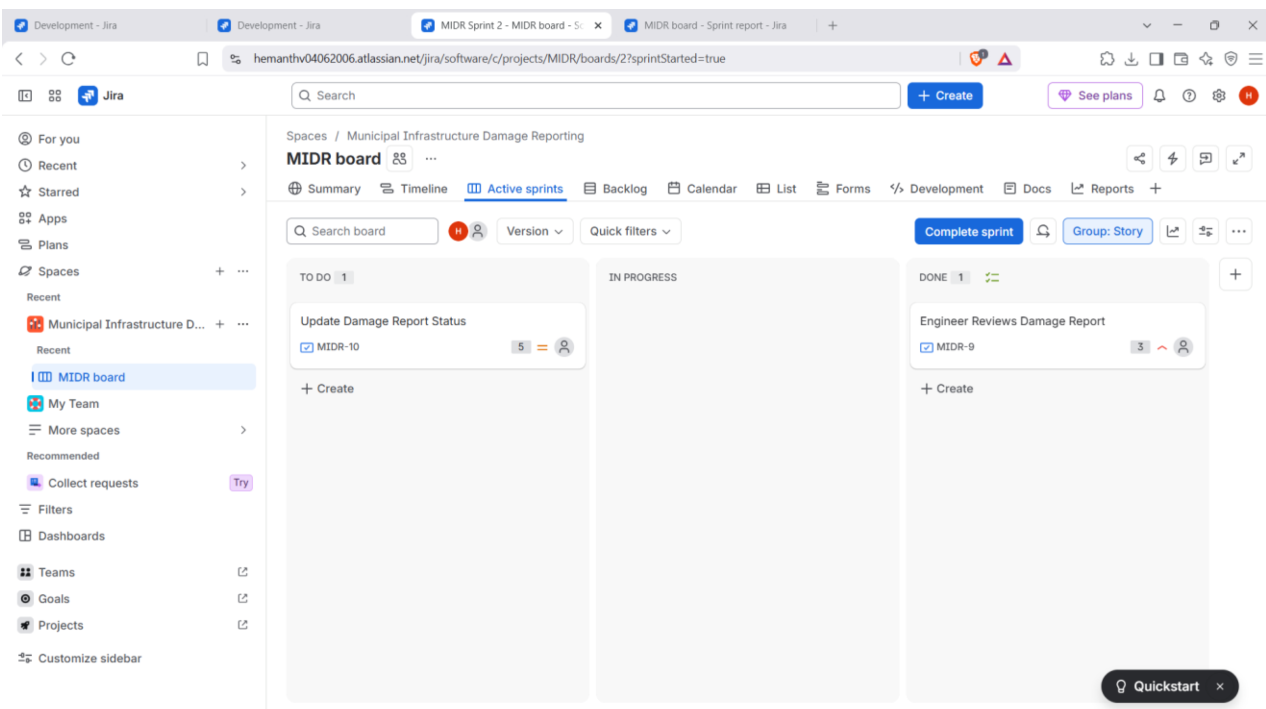
Screenshot 5 — Sprint 2 – Initial Active Sprint state



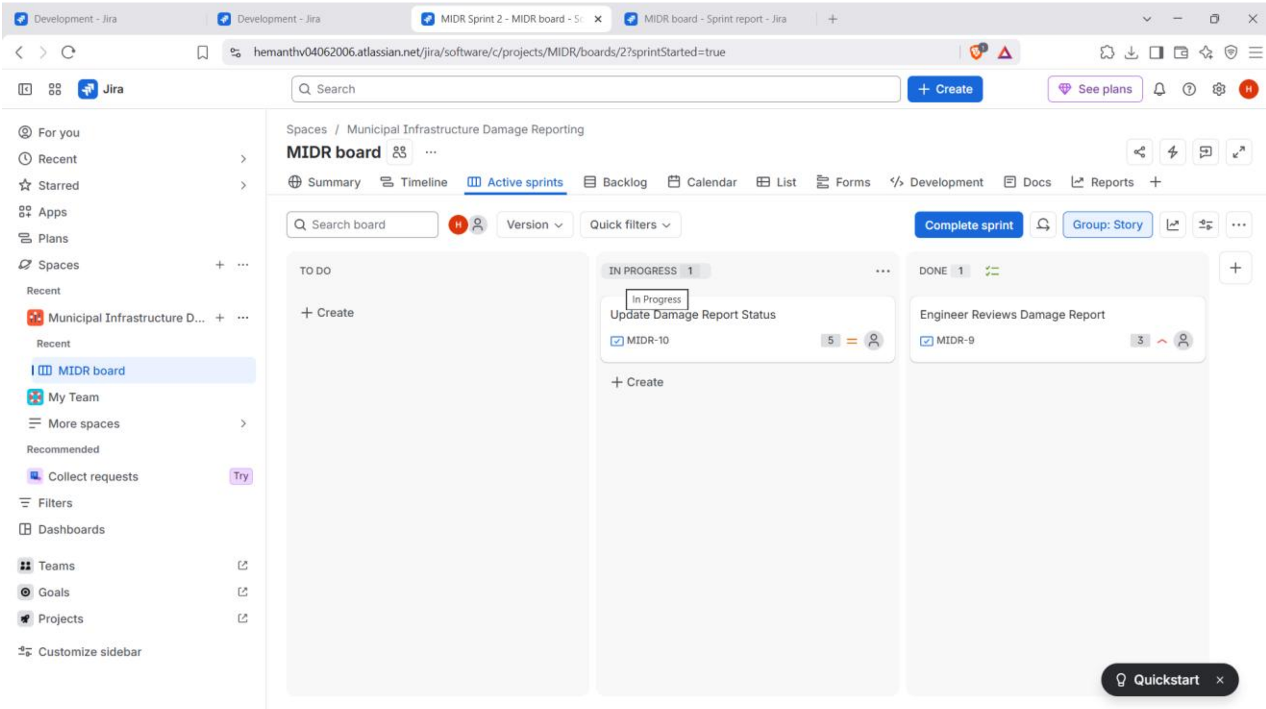
Screenshot 6 — Sprint 2 – Work in progress



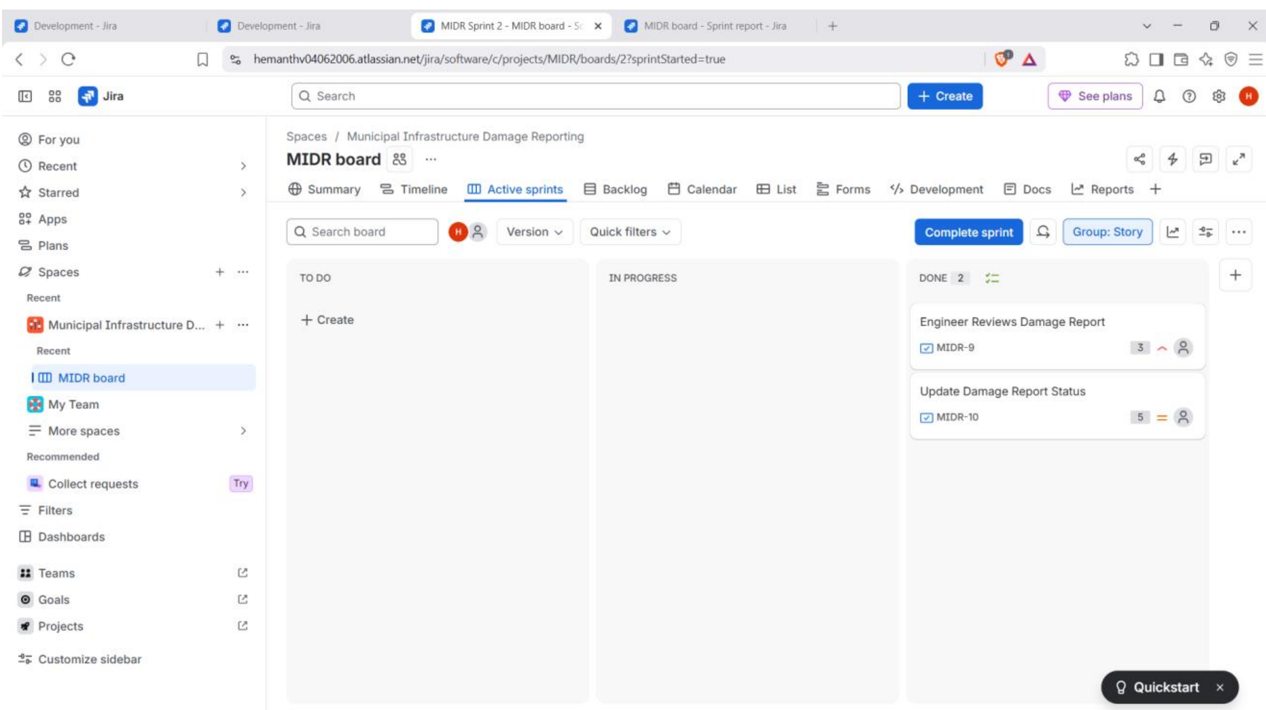
Screenshot 7 — Sprint 2 – Progress state



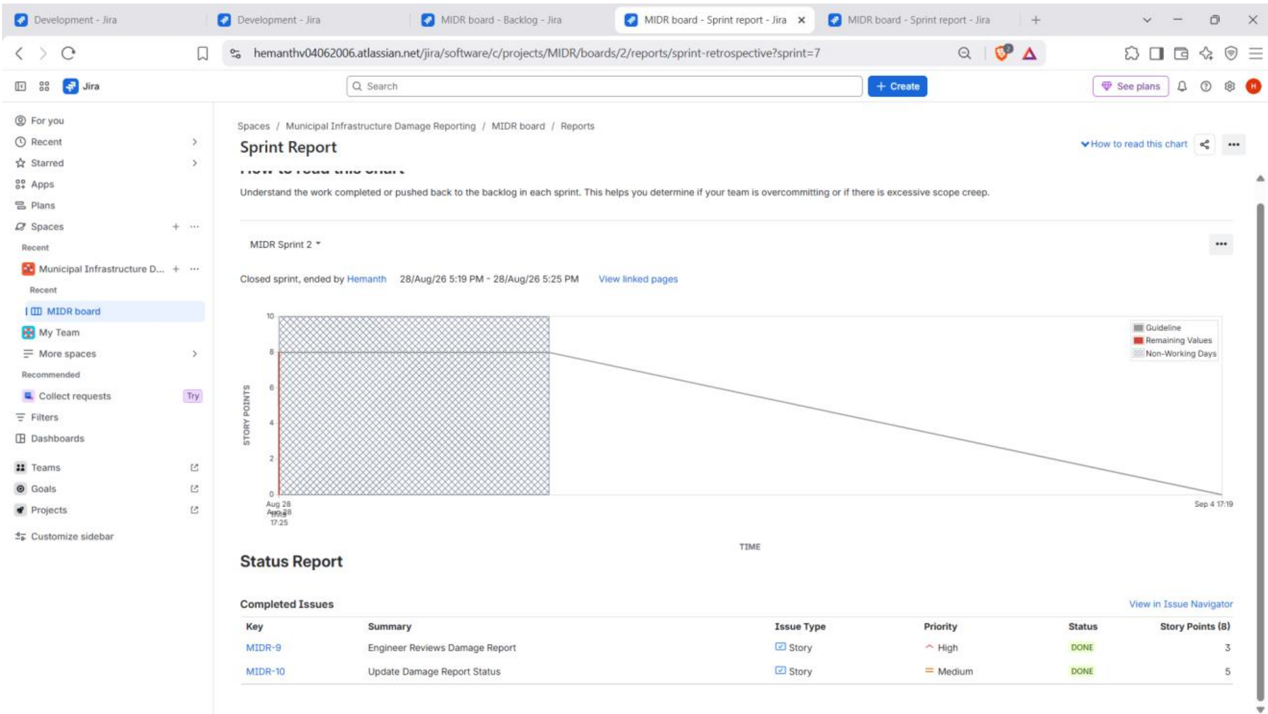
Screenshot 8 — Sprint 2 – Continued progress



Screenshot 9 — Sprint 2 – Completed work



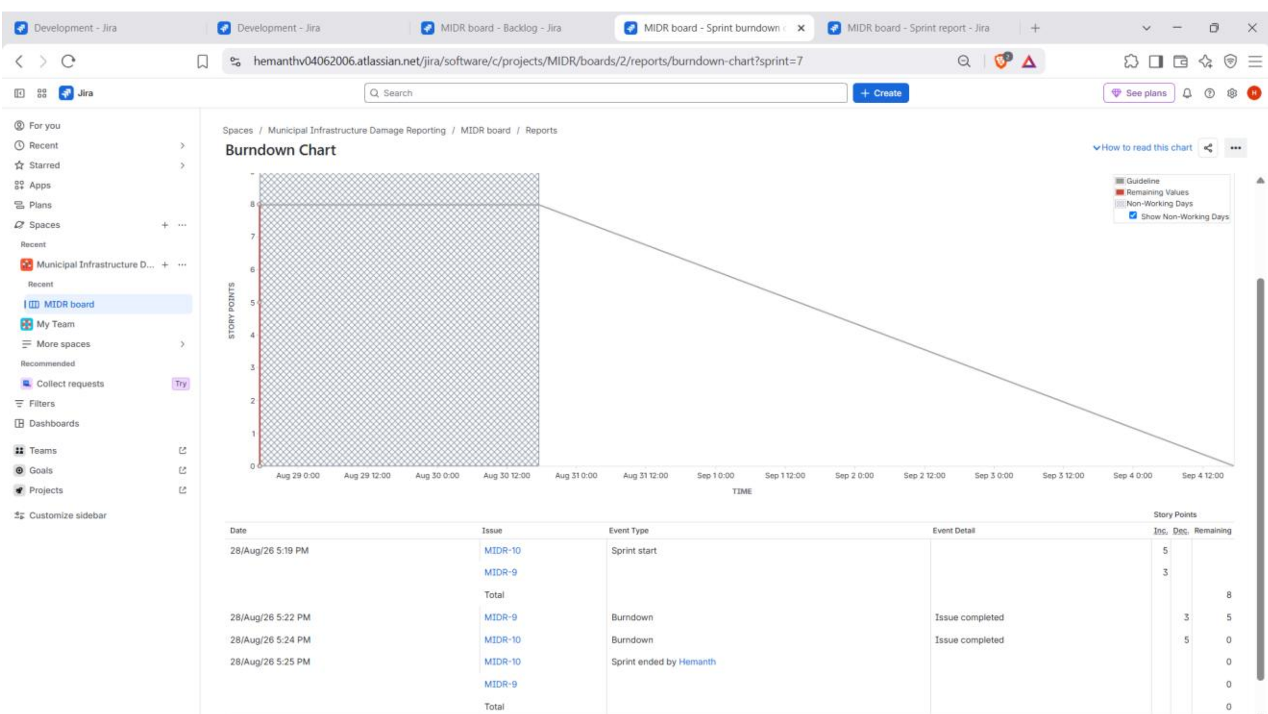
Screenshot 10 — Sprint 2 – Sprint Report



7. Burndown Chart

The Burndown Chart was generated from Jira's Reports section. It represents remaining work over the sprint and provides a visual indication of how the team progressed toward completing the planned work.

Screenshot 11 — Sprint 2 – Burndown Chart



8. Reflection and Anal

1. Did your estimations reflect the actual effort?

The estimations were reasonably aligned with the simulated effort. Sprint 1 was estimated at 21 story points and all five planned stories were completed. Sprint 2 was estimated at 8 story points and both planned stories were completed. Since this was a short simulation, the result mainly indicates that the selected estimates were sufficient for completing the planned work rather than proving long-term estimation accuracy.

2. Was your backlog well-prioritized?

Yes. The backlog was organized around the functional flow of the MIDR system, and the higher-impact reporting, photo, location, and dispatch stories were given High priority. Status-management work was also included with appropriate priority. The ordering allowed the main damage-reporting workflow to be completed before the later engineer-management work.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned well with the planned workflow. In each sprint, the selected stories were moved through To Do, In Progress, and Done. Sprint 1 completed five stories totaling 21 points, and Sprint 2 completed two stories totaling 8 points. This shows that the planned work was completed within the simulated sprint.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart provided a visual view of remaining work as the sprint progressed. It showed that the planned work was ultimately completed and therefore gave evidence that the selected Sprint 2 workload was manageable for the simulated sprint. In a real project, repeated sprint results could be used to establish a more reliable team capacity and improve future sprint planning.

9. Conclusion

This laboratory demonstrated the Agile workflow in Jira by creating and organizing Epics and User Stories, prioritizing and estimating work using story points, running two simulated sprints, moving work through the Scrum board, and analysing sprint progress using Jira's Sprint Reports and Burndown Chart. The completed evidence shows Sprint 1 with 21 story points and Sprint 2 with 8 story points, with the planned stories completed in both sprints.